

AI Exploratory Test Report

Target Application: https://formverse-explorer.lovable.app

Report Date: March 24, 2026 at 10:33

Tool Version: AI Exploratory Test Engine v2.0

Pages Analyzed: 10

Total Test Cases Generated: 58

Valid Test Cases: 58

Rejected (Hallucinated): 0

Tests Executed: 58

Passed: 57

Failed (Bugs Found): 1

2. Execution Results Summary

57 PASSED

Tests behaved correctly

1 FAILED

Bugs found in application

Bugs Found - Failed Test Cases:

Test ID	Goal	Expected (Not Met - Bug)
VT-40	Test the survey form's mode switching by changing the g	No UI errors

3. Application Coverage Summary

Metric	Value
Pages Crawled	10
Pages with Interactive Elements	10
Input Fields Identified	30
Buttons Identified	27
Navigation Links	110
Dropdowns/Selects	0
Test Cases Generated	58

Validation Pass Rate	100.0%
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4. Batch Generation Strategy

Tests were generated in 6 adversarial batches. Each batch focused on a distinct exploration category to guarantee structural diversity and maximize bug-finding potential.

Batch	Category	Focus
1	Input Torture and Field Poisoning	Edge-case values: long strings, special chars, min-1/max+1, empty required fields.
2	State Transitions and Form Abandonment	Partial fills, navigate-away-return, recalculate after change, state persistence.
3	Boundary Value Analysis	Exact min, max, zero, +1/-1 boundaries across every numeric input field.
4	Multi-Step Navigation Chaos	Fill on page A, jump to B, return to A -- does state survive?
5	Mode Switching and UI State Dances	Toggle modes mid-fill, switch tabs, check value retention after mode change.
6	Error Recovery and Validation Resilience	Submit empty, fix one field, resubmit; clear after success; recovery flow testing.

5. Test Coverage Classification

Category	Count
General Exploratory Tests	23
Negative Input Tests	12
Boundary Tests	8
Authentication Tests	5
Zero/Empty Input Tests	4
Validation Enforcement Tests	3
State Persistence Tests	3

Test cases are automatically classified based on their goals and step patterns. Categories are not mutually exclusive; each test is assigned to its primary category.

6. Validated Test Cases

58 test cases passed anti-hallucination validation. All referenced UI elements exist in the application snapshot.

Each test is marked PASSED (green) or FAILED (red) based on execution results.

VT-01 | Break registration form with long string input [PASSED]

Page: /forms/registration

Target: :r0:

Type: Boundary Tests

- 1. Navigate to '/forms/registration'
- 2. Enter '999999999' into ':r0:'
- 3. Click 'Register'

Expected: validation failure

VT-02 | Poison login form with special characters [PASSED]

Page: /forms/login

Target: :r0:

Type: Negative Input Tests

- 1. Navigate to '/forms/login'
- 2. Enter '!@#\$\$' into ':r0:'
- 3. Click 'Sign In'

Expected: error message

VT-03 | Contaminate contact form with cross-field input [PASSED]

Page: /forms/contact

Target: :r0:

Type: Negative Input Tests

- 1. Navigate to '/forms/contact'
- 2. Enter 'abc' into ':r0:'
- 3. Enter 'abc123' into ':r1:'
- 4. Click 'Send Message'

Expected: validation failure

VT-04 | Break job application form with empty required field [PASSED]

Page: /forms/job-application

Target: :r0:

Type: Validation Enforcement Tests

- 1. Navigate to '/forms/job-application'
- 2. Enter " into ':r0:'
- 3. Click 'Submit Application'

Expected: error message

VT-05 | Torture feedback form with whitespace-only input [PASSED]

Page: /forms/feedback

Target: :r0:

Type: General Exploratory Tests

- 1. Navigate to '/forms/feedback'
- 2. Enter ' ' into ':r0:'
- 3. Click 'Submit Feedback'

Expected: validation failure

VT-06 | Exceed event registration form limits [PASSED]

Page: /forms/event-registration

Target: :r4:

Type: Boundary Tests

- 1. Navigate to '/forms/event-registration'
- 2. Enter '11' into ':r4:'
- 3. Click 'Register'

Expected: error message

VT-07 | Poison survey form with invalid input [PASSED]

Page: /forms/survey

Target: :r0:

Type: Negative Input Tests

1. Navigate to '/forms/survey'
2. Enter '-1' into ':r0:'
3. Click 'Submit Survey'

*Expected: validation failure***VT-08 | Break appointment form with long string input [PASSED]**

Page: /forms/appointment

Target: :r0:

Type: Boundary Tests

1. Navigate to '/forms/appointment'
2. Enter '999999999' into ':r0:'
3. Click 'Book Appointment'

*Expected: error message***VT-09 | Contaminate registration form with alphanumerics [PASSED]**

Page: /forms/registration

Target: :r1:

Type: Negative Input Tests

1. Navigate to '/forms/registration'
2. Enter 'abc123' into ':r1:'
3. Click 'Register'

*Expected: validation failure***VT-10 | Torture login form with spaces-only input [PASSED]**

Page: /forms/login

Target: :r0:

Type: Authentication Tests

1. Navigate to '/forms/login'
2. Enter ' ' into ':r0:'
3. Click 'Sign In'

*Expected: error message***VT-11 | Test registration form with missing fields [PASSED]**

Page: /forms/registration

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/registration'
2. Enter 'test' into ':r0:'
3. Click 'Register'

*Expected: Validation error for missing fields***VT-12 | Test login form with empty fields [PASSED]**

Page: /forms/login

Target: N/A

Type: Authentication Tests

1. Navigate to '/forms/login'
2. Click 'Sign In'

*Expected: Validation error for empty fields***VT-13 | Test contact form with partial data [PASSED]**

Page: /forms/contact

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/contact'
2. Enter 'test' into ':r0:'
3. Click 'Send Message'

Expected: Validation error for missing fields

VT-14 | Test job application form with valid data and immediate resu [PASSED]

Page: /forms/job-application

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/job-application'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Enter 'test' into ':r3:'
6. Click 'Submit Application'
7. Click 'Submit Application'

Expected: Validation error for duplicate submission

VT-15 | Test feedback form with empty fields [PASSED]

Page: /forms/feedback

Target: N/A

Type: Zero/Empty Input Tests

1. Navigate to '/forms/feedback'
2. Click 'Submit Feedback'

Expected: Validation error for empty fields

VT-16 | Test survey form with missing fields [PASSED]

Page: /forms/survey

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/survey'
2. Enter 'test' into ':r0:'
3. Click 'Submit Survey'

Expected: Validation error for missing fields

VT-17 | Test appointment form with partial data [PASSED]

Page: /forms/appointment

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/appointment'
2. Enter 'test' into ':r0:'
3. Click 'Book Appointment'

Expected: Validation error for missing fields

VT-18 | Test registration form with valid data and modification of a [PASSED]

Page: /forms/registration

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/registration'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Enter 'test' into ':r3:'
6. Enter 'test' into ':r4:'
7. Click 'Register'
8. Enter 'new test' into ':r0:'
9. Click 'Register'

Expected: Validation error for changed data

VT-19 | Test login form with valid data and immediate resubmission [PASSED]

Page: /forms/login

Target: :r0:

Type: Authentication Tests

1. Navigate to '/forms/login'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Click 'Sign In'
5. Click 'Sign In'

*Expected: Validation error for duplicate submission***VT-20 | Break registration with minimum allowed value [PASSED]**

Page: /forms/registration

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/registration'
2. Enter '1' into ':r0:'
3. Click 'Register'

*Expected: Validation failure***VT-21 | Break login with maximum allowed value [PASSED]**

Page: /forms/login

Target: :r0:

Type: Boundary Tests

1. Navigate to '/forms/login'
2. Enter '9999999999' into ':r0:'
3. Click 'Sign In'

*Expected: Error message***VT-22 | Break contact with one below minimum allowed value [PASSED]**

Page: /forms/contact

Target: :r0:

Type: Boundary Tests

1. Navigate to '/forms/contact'
2. Enter '0' into ':r0:'
3. Click 'Send Message'

*Expected: Validation failure***VT-23 | Break job-application with one above maximum allowed value [PASSED]**

Page: /forms/job-application

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/job-application'
2. Enter '11' into ':r0:'
3. Click 'Submit Application'

*Expected: Error message***VT-24 | Break feedback with zero where non-zero is expected [PASSED]**

Page: /forms/feedback

Target: :r0:

Type: Boundary Tests

1. Navigate to '/forms/feedback'
2. Enter '0' into ':r0:'
3. Click 'Submit Feedback'

Expected: Validation failure

VT-25 | Break event-registration with very large number [PASSED]

Page: /forms/event-registration

Target: :r1:

Type: Boundary Tests

1. Navigate to '/forms/event-registration'
2. Enter '9999999999' into ':r1:'
3. Click 'Register'

*Expected: Error message***VT-26 | Break survey with very small decimal [PASSED]**

Page: /forms/survey

Target: :r0:

Type: Boundary Tests

1. Navigate to '/forms/survey'
2. Enter '0.0001' into ':r0:'
3. Click 'Submit Survey'

*Expected: Validation failure***VT-27 | Break appointment with invalid input [PASSED]**

Page: /forms/appointment

Target: :r0:

Type: Negative Input Tests

1. Navigate to '/forms/appointment'
2. Enter 'abc' into ':r0:'
3. Click 'Book Appointment'

*Expected: Error message***VT-28 | Break registration with empty input [PASSED]**

Page: /forms/registration

Target: :r1:

Type: Zero/Empty Input Tests

1. Navigate to '/forms/registration'
2. Enter "" into ':r1:'
3. Click 'Register'

*Expected: Validation failure***VT-29 | Test rapid submission of registration form [PASSED]**

Page: /forms/registration

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/registration'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Enter 'test' into ':r3:'
6. Enter 'test' into ':r4:'
7. Click 'Register'
8. Click 'Register' again

*Expected: Validation failure due to duplicate submission***VT-30 | Test submission of partially filled login form [PASSED]**

Page: /forms/login

Target: :r0:

Type: Authentication Tests

1. Navigate to '/forms/login'
2. Enter 'test' into ':r0:'
3. Click 'Sign In'

Expected: Error message for missing ':r1:' field

VT-31 | Test rapid filling of contact form fields [PASSED]

Page: /forms/contact

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/contact'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Click 'Send Message'

*Expected: No UI errors or crashes***VT-32 | Test submission of job application form with empty fields [PASSED]**

Page: /forms/job-application

Target: N/A

Type: Zero/Empty Input Tests

1. Navigate to '/forms/job-application'
2. Click 'Submit Application'

*Expected: Error message for missing required fields***VT-33 | Test submission of feedback form with valid input [PASSED]**

Page: /forms/feedback

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/feedback'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Click 'Submit Feedback'

*Expected: No UI errors or crashes***VT-34 | Test submission of event registration form with invalid input [PASSED]**

Page: /forms/event-registration

Target: :r0:

Type: Negative Input Tests

1. Navigate to '/forms/event-registration'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Enter '11' into ':r4:'
6. Click 'Register'

*Expected: Error message for ':r4:' field exceeding maximum value***VT-35 | Test submission of survey form with valid input [PASSED]**

Page: /forms/survey

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/survey'
2. Enter '50' into ':r0:'
3. Select 'option' from 'gender'
4. Enter 'test' into 'q1'
5. Enter 'test' into 'q2'
6. Click 'Submit Survey'

Expected: No UI errors or crashes

VT-36 | Test submission of appointment form with empty fields [PASSED]

Page: /forms/appointment

Target: N/A

Type: Zero/Empty Input Tests

1. Navigate to '/forms/appointment'
2. Click 'Book Appointment'

*Expected: Error message for missing required fields***VT-37 | Test rapid submission of registration form with different va [PASSED]**

Page: /forms/registration

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/registration'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Enter 'test' into ':r3:'
6. Enter 'test' into ':r4:'
7. Click 'Register'
8. Enter 'new' into ':r0:'
9. Enter 'new' into ':r1:'
10. Enter 'new' into ':r2:'
11. Enter 'new' into ':r3:'
12. Enter 'new' into ':r4:'
13. Click 'Register'

*Expected: No UI errors or crashes***VT-38 | Test submission of contact form with rapid input [PASSED]**

Page: /forms/contact

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/contact'
2. Enter 'test' into ':r0:'
3. Enter 'test' into ':r1:'
4. Enter 'test' into ':r2:'
5. Click 'Send Message'
6. Enter 'new' into ':r0:'
7. Enter 'new' into ':r1:'
8. Enter 'new' into ':r2:'
9. Click 'Send Message'

*Expected: No UI errors or crashes***VT-39 | Break the job application form by rapidly switching between [PASSED]**

Page: /forms/job-application

Target: N/A

Type: Negative Input Tests

1. Navigate to '/forms/job-application'
2. Click 'JavaScript'
3. Click 'TypeScript'
4. Click 'React'
5. Click 'Submit Application'

Expected: validation failure

VT-40 | Test the survey form's mode switching by changing the gender [FAILED - BUG FOUND]

Page: /forms/survey

Target: gender

Type: General Exploratory Tests

1. Navigate to '/forms/survey'
2. Enter 'Male' into 'gender'
3. Click 'gender'
4. Enter '30' into ':r0:'
5. Click 'Submit Survey'

*Expected: No UI errors***!! Application did not meet expected behavior - potential bug detected.****VT-41 | Stress the event registration form by rapidly switching betw [PASSED]**

Page: /forms/event-registration

Target: :r4:

Type: General Exploratory Tests

1. Navigate to '/forms/event-registration'
2. Enter '5' into ':r4:'
3. Click 'Register'
4. Click 'Dismiss'
5. Click 'Register'

*Expected: error message***VT-42 | Break the appointment form by clicking the 'Pick a date' but [PASSED]**

Page: /forms/appointment

Target: N/A

Type: General Exploratory Tests

1. Navigate to '/forms/appointment'
2. Click 'Pick a date'
3. Click 'Pick a date'
4. Click 'Pick a date'
5. Click 'Book Appointment'

*Expected: validation failure***VT-43 | Test the feedback form's mode switching by submitting the fo [PASSED]**

Page: /forms/feedback

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/feedback'
2. Enter 'Test feedback' into ':r0:'
3. Enter 'Test feedback' into ':r1:'
4. Click 'Submit Feedback'
5. Click 'Dismiss'

*Expected: No UI errors***VT-44 | Stress the registration form by rapidly switching between re [PASSED]**

Page: /forms/registration

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/registration'
2. Enter 'Test' into ':r0:'
3. Click 'Register'
4. Click 'Dismiss'
5. Click 'Register'

Expected: error message

VT-45 | Break the login form by clicking the 'Sign In' button multip [PASSED]

Page: /forms/login

Target: N/A

Type: Authentication Tests

1. Navigate to '/forms/login'
2. Click 'Sign In'
3. Click 'Sign In'
4. Click 'Sign In'
5. Click 'Dismiss'

*Expected: validation failure***VT-46 | Test the contact form's mode switching by submitting the for [PASSED]**

Page: /forms/contact

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/contact'
2. Enter 'Test' into ':r0:'
3. Enter 'Test' into ':r1:'
4. Enter 'Test' into ':r2:'
5. Click 'Send Message'

*Expected: No UI errors***VT-47 | Stress the job application form by rapidly switching between [PASSED]**

Page: /forms/job-application

Target: N/A

Type: Negative Input Tests

1. Navigate to '/forms/job-application'
2. Click 'JavaScript'
3. Click 'TypeScript'
4. Click 'React'
5. Click 'Node.js'
6. Click 'Submit Application'

*Expected: error message***VT-48 | Break the survey form by clicking the 'Submit Survey' button [PASSED]**

Page: /forms/survey

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/survey'
2. Enter '30' into ':r0:'
3. Click 'Submit Survey'
4. Click 'Submit Survey'
5. Click 'Submit Survey'

*Expected: validation failure***VT-49 | Trigger error by submitting empty form and then recover by f [PASSED]**

Page: /forms/registration

Target: :r0:

Type: Validation Enforcement Tests

1. Navigate to '/forms/registration'
2. Click 'Register'
3. Enter 'valid_value' into ':r0:'
4. Click 'Register'

Expected: Error message clears after filling one field

VT-50 | Test error recovery after invalid input [PASSED]

Page: /forms/event-registration

Target: :r4:

Type: Negative Input Tests

1. Navigate to '/forms/event-registration'
2. Enter '11' into ':r4:'
3. Click 'Register'
4. Enter '5' into ':r4:'
5. Click 'Register'

*Expected: Error message clears after correcting invalid input***VT-51 | Trigger calculation error and then provide valid input [PASSED]**

Page: /forms/survey

Target: :r0:

Type: General Exploratory Tests

1. Navigate to '/forms/survey'
2. Enter '121' into ':r0:'
3. Click 'Submit Survey'
4. Enter '60' into ':r0:'
5. Click 'Submit Survey'

*Expected: Error message clears after providing valid input***VT-52 | Test error handling after clearing fields [PASSED]**

Page: /forms/registration

Target: :r0:

Type: State Persistence Tests

1. Navigate to '/forms/registration'
2. Enter 'valid_value' into ':r0:'
3. Enter 'valid_value' into ':r1:'
4. Click 'Register'
5. Click 'Dismiss'
6. Click 'Register'

*Expected: Error message appears after submitting empty form***VT-53 | Test error recovery after clicking Clear/Reset [PASSED]**

Page: /forms/contact

Target: :r0:

Type: State Persistence Tests

1. Navigate to '/forms/contact'
2. Enter 'valid_value' into ':r0:'
3. Enter 'valid_value' into ':r1:'
4. Click 'Dismiss'
5. Click 'Send Message'

*Expected: Error message appears after submitting empty form***VT-54 | Trigger error by submitting form with invalid value [PASSED]**

Page: /forms/job-application

Target: :r0:

Type: Negative Input Tests

1. Navigate to '/forms/job-application'
2. Enter 'invalid_value' into ':r0:'
3. Click 'Submit Application'
4. Enter 'valid_value' into ':r0:'
5. Click 'Submit Application'

Expected: Error message clears after correcting invalid input

VT-55 | Test error handling after providing valid data and then clear [PASSED]

Page: /forms/feedback

Target: :r0:

Type: State Persistence Tests

1. Navigate to '/forms/feedback'
2. Enter 'valid_value' into ':r0:'
3. Enter 'valid_value' into ':r1:'
4. Click 'Dismiss'
5. Click 'Submit Feedback'

*Expected: Error message appears after submitting empty form***VT-56 | Trigger error by submitting form with missing required field [PASSED]**

Page: /forms/appointment

Target: :r0:

Type: Validation Enforcement Tests

1. Navigate to '/forms/appointment'
2. Click 'Book Appointment'
3. Enter 'valid_value' into ':r0:'
4. Click 'Book Appointment'

*Expected: Error message clears after filling required fields***VT-57 | Test error recovery after correcting invalid input [PASSED]**

Page: /forms/survey

Target: q1

Type: Negative Input Tests

1. Navigate to '/forms/survey'
2. Enter 'invalid_value' into 'q1'
3. Click 'Submit Survey'
4. Enter 'valid_value' into 'q1'
5. Click 'Submit Survey'

*Expected: Error message clears after correcting invalid input***VT-58 | Trigger error by submitting form with invalid format [PASSED]**

Page: /forms/event-registration

Target: :r0:

Type: Negative Input Tests

1. Navigate to '/forms/event-registration'
2. Enter 'invalid_format' into ':r0:'
3. Click 'Register'
4. Enter 'valid_format' into ':r0:'
5. Click 'Register'

*Expected: Error message clears after correcting invalid format***7. Structural Validation Confirmation**

All 58 test cases were validated against the UI exploration snapshot. The validation engine verified that:

- ✓ All referenced UI element labels exist in the application snapshot.
- ✓ All referenced page paths exist in the crawled page inventory.
- ✓ No hallucinated or invented element labels were detected.
- ✓ Each test case has a valid structure: ID, goal, steps (min 2), and expected result.

8. Appendix

Snapshot Statistics

Metric	Value
Snapshot File	ai_exploration_snapshot.json
Snapshot Size	42429 bytes
Total Pages in Snapshot	10
Total Input Fields	30
Total Buttons	27
Total Navigation Links	110

Pipeline Configuration

Parameter	Value
AI Model	llama-3.3-70b-versatile (Grok)
Generation Batches	6
Tests per Batch	~10
Batch Delay	30 seconds
Validation Engine	3-Layer Anti-Hallucination
Step Generator	Deterministic (Template-Based)

Output Files Reference

File	Path
Raw Test Cases	data/generated_test_cases.json
Validated Test Cases	data/validated_test_cases.json
Rejected Test Cases	data/rejected_test_cases.json
Gauge Specification	specs/ai_exploration.spec
Step Implementations	step_impl/step_implementation.py
This Report	reports/test_report.pdf